

Ignoring air resistance, a falling object accelerates at 9.8 m/s^2 . An object is dropped from a height of 20 m. How fast is it moving when it hits the ground?

What is the general form for an antiderivative of

$$f(x) = \frac{1}{x} ?$$

Theorem: Let $f(x)$ be a function defined on an interval, and let $F(x)$ be any antiderivative of f on that interval. Then all antiderivatives of f on that interval are of the form.

$$G(x) = F(x) + C$$

where C is a real constant.

On the interval $(0, \infty)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln x + C$$

On the interval $(-\infty, 0)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln(-x) + D$$

You will often see these combined as

$$\ln |x| + C$$

This should be understood as the form of all antiderivatives on $(0, \infty)$, or all antiderivatives on $(-\infty, 0)$, but not all antiderivatives defined on $(-\infty, 0) \cup (0, \infty)$ are of this form.

The Area Problem (5.1)

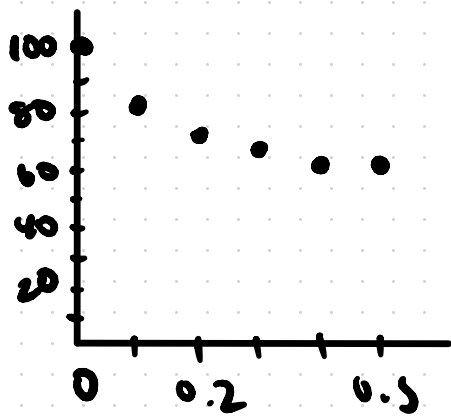
Zeno's arrow flies through the air in a line. Its velocity at various times is given below.

Time (s)	0	0.1	0.2	0.3	0.4	0.5
Velocity (m/s)	100	100	95	90	80	70

About how far has it traveled after half a second?

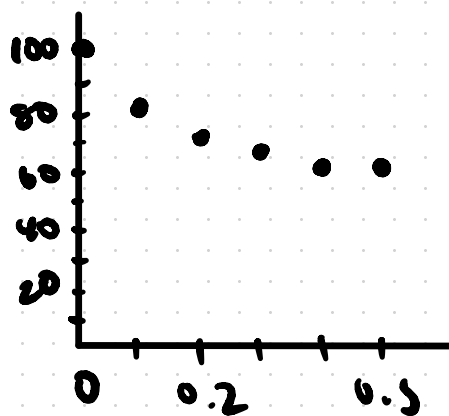
Time (s)	0	0.1	0.2	0.3	0.4	0.5
Velocity (m/s)	100	80	70	65	60	60

Velocity (m/s)

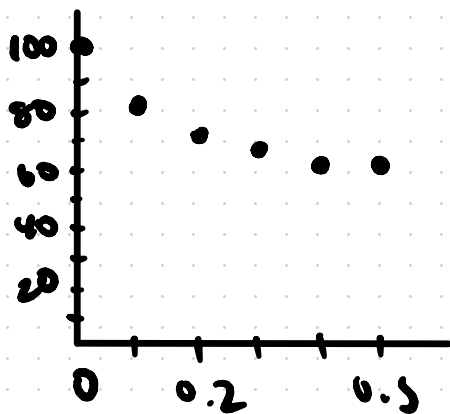


Time(s)

Velocity (m/s)



Time(s)



Time(s)

What is the area of a circle with radius 1?

What is the area of the region bounded by the x -axis, y -axis, and the graph of $y = 1 - x^2$?